

Defect Log

Defects found and fixed while adding the test suite to this codebase. All were found by actually running the app/tests, not invented.

DEF-001

Field	Value
Title	Admin sidebar navigation links never render, even for admin users
Description	base.html checked <code>current_user.is_admin</code> , but User never defined that attribute (only <code>role</code>). dashboard.html's own quick-actions panel correctly checked <code>role == "admin"</code> , so the two templates were inconsistent - admins saw their dashboard quick-actions but never the sidebar links.
Severity	Medium (UI-only; routes were still protected by <code>admin_required</code> , so no security impact, but admins effectively couldn't navigate to admin screens from the sidebar)
Priority	High
Status	Closed
Detected by	Manual reproduction (curl as admin, grepped rendered HTML for expected nav text); regression-tested by <code>test_it109_admin_sidebar_link_visible_for_admin</code>
Root cause	app/models.py: User had no <code>is_admin</code> property
Fix	Added <code>is_admin</code> property to User (<code>role == "admin"</code>)
Retest	Pass - IT-109/IT-110 confirm the sidebar now shows/hides correctly for admin/student

DEF-002

Field	Value
Title	<code>approve_booking</code> always fails when a booking uses all remaining capacity
Description	The availability re-check inside <code>approve_booking</code> calls <code>calculate_reserved_quantity</code> , which counted <i>all</i> REQUESTED/APPROVED/ACTIVE/LATE bookings overlapping the period - including the very booking being approved, since it is still REQUESTED at that point. This double-counts the booking against itself, so approval fails whenever the booking's own quantity equals the remaining stock (the single most common case for low-quantity items).
Severity	High - this breaks the core approval workflow for any equipment item close to its capacity
Priority	High
Status	Closed
Detected by	<code>test_it305_full_lifecycle_approve_activate_return</code> (integration test) - first run failed at the approve step with a real, reproducible <code>ValueError</code>
Root cause	<code>get_overlapping_bookings / calculate_reserved_quantity / check_availability</code> had no way to exclude a specific booking ID from the reserved-quantity count
Fix	Added an <code>exclude_booking_id</code> parameter threaded through <code>get_overlapping_bookings -> calculate_reserved_quantity -> check_availability</code> , and <code>approve_booking</code> now passes <code>exclude_booking_id=booking.id</code>
Retest	Pass - full suite (115/115) including the conflict test (<code>test_ut015_approve_rechecks_availability_and_rejects_conflict</code>), which still correctly rejects genuine conflicts from other

DEF-003

Field	Value
Title	Dead app/services.py module conflicting with app/services/ package
Description	Both a services.py file and a services/ package existed in app/. Routes only ever imported from the package; the loose file was orphaned dead code and a latent import-ambiguity risk.
Severity	Low
Priority	Medium
Status	Closed
Fix	Removed app/services.py and the unused legacy compatibility shims in services/__init__.py after confirming (via grep) nothing referenced them

DEF-004

Field	Value
Title	Equipment form exposed a non-functional “Available Quantity” field
Description	equipment_form.html let an admin type a value into an “Available Quantity” input, but add_equipment/edit_equipment never read that field from the submitted form - the server always computed it itself. The field was misleading UI with no effect.
Severity	Low
Priority	Low
Status	Closed
Fix	Removed the field from the template; confirmed via test_it201_admin_can_add_equipment that available_quantity is still computed correctly server-side

Missing feature (not a defect - closed as an enhancement)

The brief requires REQUESTED->CANCELLED and APPROVED->CANCELLED as valid state transitions with a cancellation deadline. This codebase had no cancellation feature at all - no route, no service function, CANCELLED absent from the transition table. Added: cancel_booking() (owner-or-admin permission check, deadline enforcement), the /bookings/<id>/cancel route, and a Cancel button in the UI. Covered by UT-028 through UT-033, IT-307 through IT-309, and E2E-008/E2E-011.

Summary

Status	Count
Closed	4 defects + 1 missing feature added
Open	0
Severity	Count
High	1 (DEF-002)
Medium	1 (DEF-001)
Low	2 (DEF-003, DEF-004)

Known tech debt (not fixed, flagged for transparency)

datetime.utcnow() is used throughout routes.py and booking_service.py; functionally correct on Python 3.12 but raises a DeprecationWarning (scheduled for removal in a future Python version). Not fixed here since it doesn't affect any test result or business behavior.

